

Change record — P1 Core Infrastructure Build

Retroactive change record, written to the standard Halden uses for every change (feeds the change log and the P10 governance dashboard). In a small company the same person often proposes and approves; that separation is documented honestly rather than pretended.

Field	Value
Change ID	CHG-2026-001
Title	P1 — Core infrastructure build: AD DS, DNS, DHCP, file services, GPO baseline, Linux integration
Raised by	IT (Muhammad Ibrahim Akmal)
Approved by	Managing Director (<i>business owner of the fictional Halden; see note on the lab</i>)
Date raised	2026-09-29
Planned implementation	2026-10-02 onwards, phase by phase (Phase 0 → Phase 8)
Category / risk	Infrastructure — High risk (Tier 0: authentication, DNS, DHCP)
Affected services	Logon, DNS, DHCP, all file shares, Group Policy, Linux SSH access
Affected systems	HOST01 (Hyper-V), FW01, DC01, DC02, FS01, LNX01, WS01, WS02

1. Description of the change

Replace the single aging server with a redundant, documented Windows Server 2025 foundation: two replicating domain controllers with AD-integrated DNS, a DHCP failover pair, a dedicated file server with DFS namespace, access-based enumeration, FSRM quotas and file screens and shadow copies, a tiered Group Policy baseline, and one Ubuntu server joined to the domain with group-controlled SSH and sudo. All of it is created by idempotent, logged scripts kept in the repository, plus an as-built document, a permission matrix and user runbooks.

2. Reason for the change

The current single server is a single point of failure for logon, addressing and files, and all 85 staff hold Full Control on the shared drive so a single mistake or a ransomware infection can destroy any department's data. There is also no written answer to "who can read Finance data" and no self-service file restore. The change removes the single point of failure, applies least privilege by role, and produces the documentation and evidence that the business currently lacks.

3. Impact and risk

Area	Impact	Mitigation
Users	New drive letters appear; a screen lock is enforced; folders that a user has no right to open are hidden	1-page brief to all staff in advance; Service Desk briefed; ticket route for missing access

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Logon / DNS / DHCP	Outage would stop the whole company	Work done out of hours, phase by phase; DC02 and DHCP failover tested by powering DC01 off
Data	New permission model could lock a user out of a folder they need	Access approved by department heads first; permission matrix kept; Administrators retain recovery access
Access control	A mistake could grant too much access	Only DL_ groups ever appear on ACLs; automated ACL audit must report 0 violations

4. Implementation plan (phases)

Phase	Work	Window	Verification
0	Design document (IP plan, naming, OU tree, DNS/DHCP design)	2026-10-02	Design reviewed against the plan
1	DC01 promoted; DNS zones, forwarders, scavenging, time source	Out of hours	<code>dcdiag /q</code> clean; DNS records resolve
2	DC02 promoted; sites/subnets; replication verified	Out of hours	<code>repadmin /replsummary</code> 0 fails
3	DHCP scope, options, 50/50 failover, name protection	Out of hours	Failover state Normal; client renews with DC01 off
4	OU tree, AGDLP groups, 85 users imported from CSV	Out of hours	Import log; idempotent second run
5	FS01 shares, ACLs, DFS-N, ABE, FSRM, shadow copies	Out of hours	ACL audit 0 violations
6	GPO baseline + Central Store + GPO backup	Out of hours	<code>gpresult /h</code> on WS01
7	LNK01 joined; AD-based SSH and sudo	Out of hours	Denied for non-members; sudo for members
8	As-built doc, diagram, runbooks	Same evening	Documents produced from live output

5. Test plan (acceptance)

1. Power off DC01 — logons, DNS resolution and DHCP leases continue via DC02.
2. Finance user opens the company share — the HR folder is not visible.
3. Sales user copies an `.exe` into the Sales share — blocked by the file screen.
4. Delete a file — restore it through "Previous Versions" without IT.
5. Non-IT domain user connects to LNK01 by SSH — denied.
6. Re-run the user import script — no duplicates, no errors (idempotent).

6. Backout plan

Each phase is snapshotted before it runs (`snap-p1-ph<N>-before`). Backout for any phase is to power off the affected VM, revert the snapshot and re-run the previous phase's scripts, which are idempotent. Domain controllers are fixed forward where possible: reverting a DC snapshot can disturb replication, so it is only used as a last resort and is recorded in `DECISIONS.md` .

7. Post-implementation review

To be completed when the last phase is verified: actual completion date, acceptance test results, anything that went wrong, and whether any phase overran. Results are recorded in `projects/p01-core-infrastructure/README.md` with a source file for every number.

Note: this is a home-lab change record for a fictional company (Halden Distribution Ltd.). The technical content, scripts and test results are real; the business approval line represents the lab owner's decision, not a real customer sign-off.